Task 4 – AWS Data Engineering Services Workflow

Overview of AWS Architecture

To process the e-commerce CSV files in a scalable cloud environment, a serverless modern data engineering architecture can be established using core AWS analytics and storage services.

Component Services & Their Roles

1. Amazon S3 (Simple Storage Service)

- **Purpose:** Acts as the central data lake storage layer.
- **Implementation:**
    - **Raw Layer (`s3://company-bucket/raw/`):** Stores incoming CSV files (`customers.csv`, `products.csv`, `orders.csv`) exactly as received.
    - **Processed/Curated Layer (`s3://company-bucket/processed/`):** Stores cleaned outputs and transformed datasets like `final_orders.csv`.

2. AWS Glue

- **Purpose:** Fully managed serverless ETL (Extract, Transform, Load) data integration service.
- **Implementation:**
    - **Glue Crawlers:** Automatically scan raw CSV files in Amazon S3 to infer schemas and populate the data catalog.
    - **Glue ETL Jobs (PySpark/Python):** Write data transformation scripts to remove duplicates, handle missing/orphan keys, join datasets, calculate `total_amount`, and write clean results back to S3.

3. AWS Glue Data Catalog

- **Purpose:** Serves as the central metadata repository for all data assets.
- **Implementation:** Stores database and table definitions pointing to S3 file locations, making structured and unstructured datasets discoverable for query engines.

4. Amazon Athena

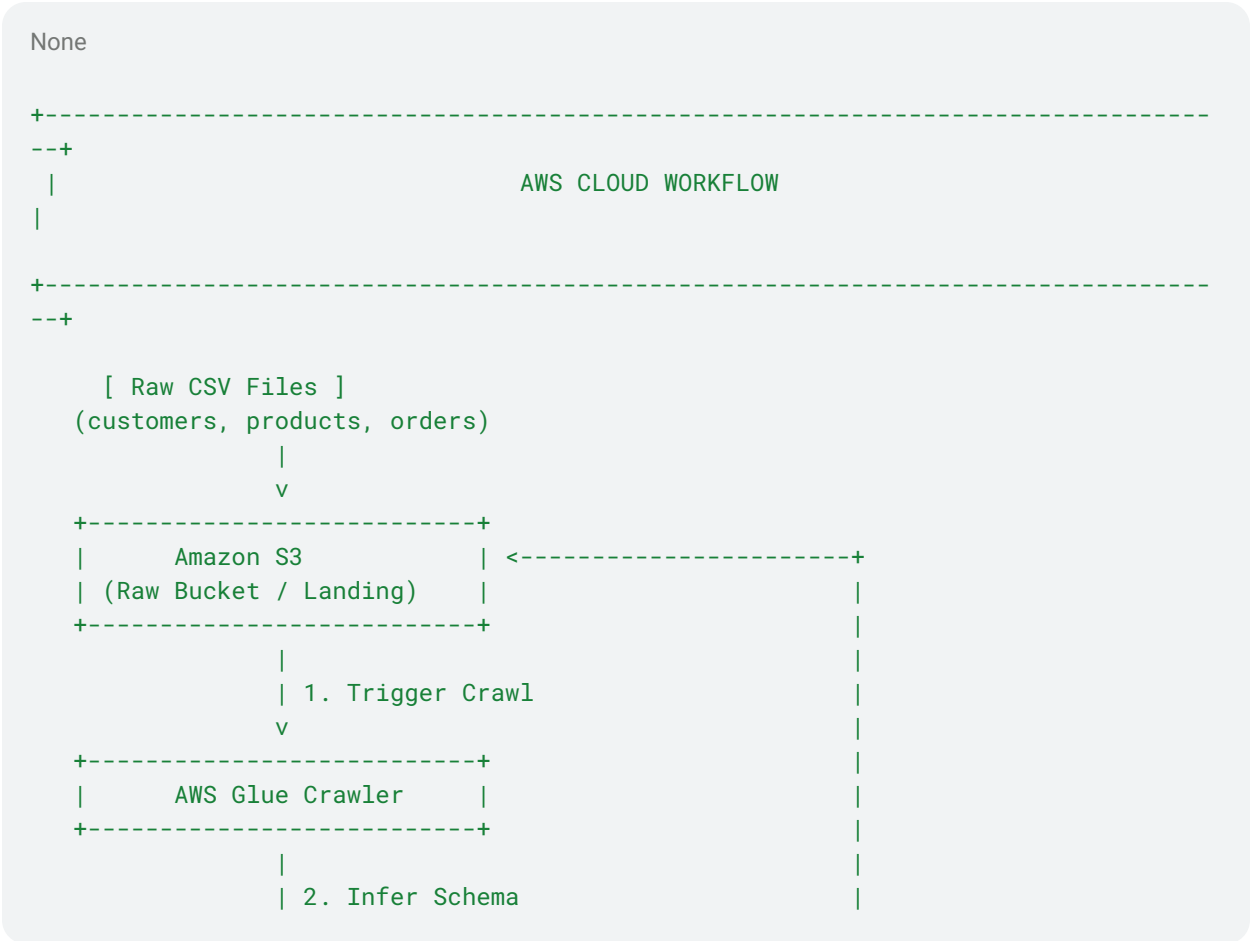
- **Purpose:** Serverless, interactive query service that analyzes data directly in Amazon S3 using standard SQL.
- **Implementation:** Allows data engineers and business analysts to run SQL queries directly on the Glue Data Catalog tables without needing to manage servers or data warehouses.

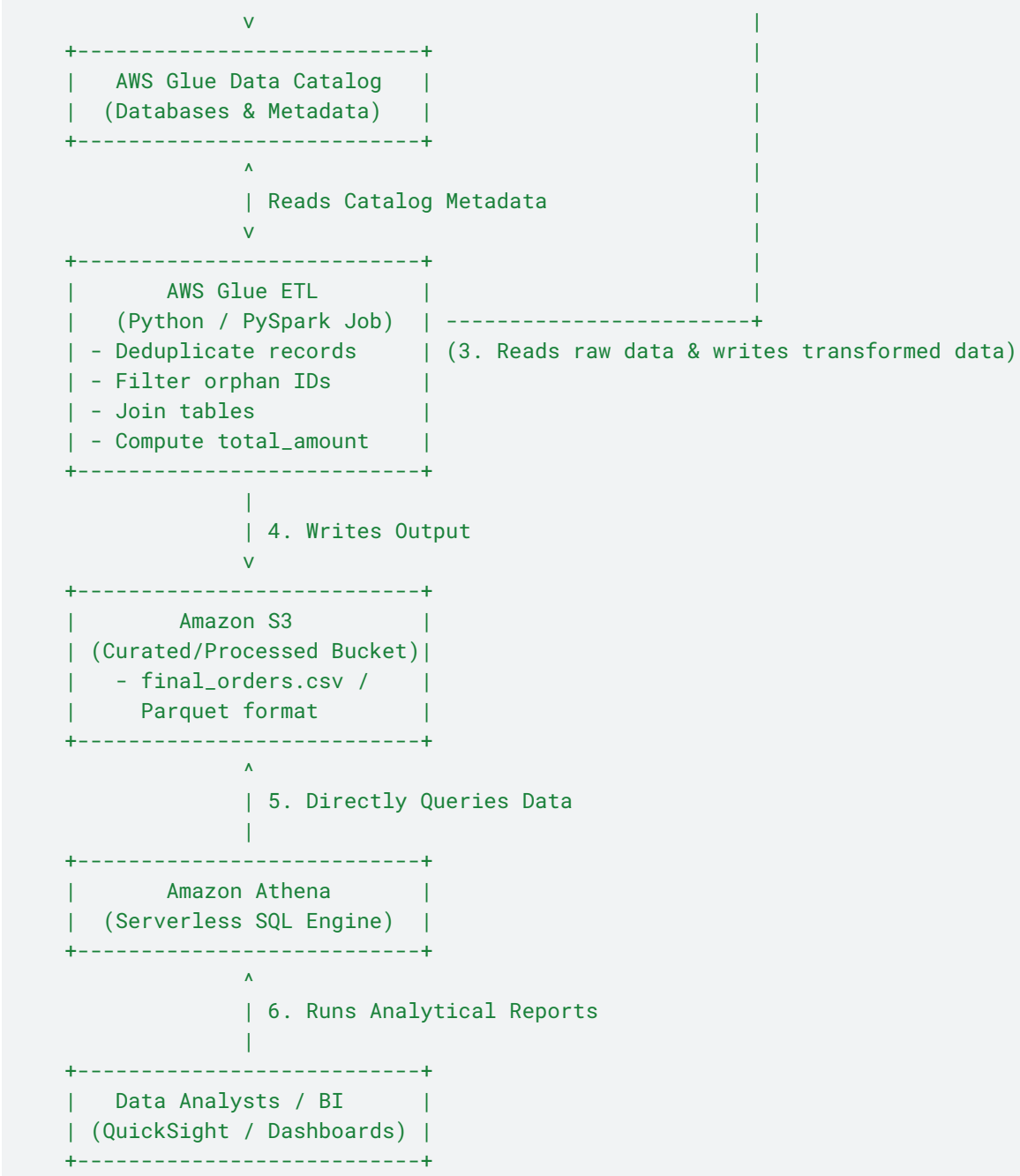
5. AWS IAM (Identity and Access Management)

- **Purpose:** Securely manages access and permissions across all AWS resources.
- **Implementation:** Implements the principle of least privilege—granting AWS Glue and Amazon Athena specific, restricted permissions to read from raw S3 buckets and write to processed buckets.

Architecture & Workflow Diagram

None





* Security & Governance:
[AWS IAM] enforces least-privilege role-based access across S3, Glue, and Athena.

Workflow Sequence

1. **Ingestion:** CSV files are uploaded into the **Amazon S3** raw bucket.
2. **Cataloging:** An **AWS Glue Crawler** runs against the raw bucket, updating the **AWS Glue Data Catalog**.
3. **Transformation:** An **AWS Glue Python Shell or Spark Job** reads the raw files, cleans records, performs joins, computes order values, and writes the clean dataset back to the S3 processed bucket.
4. **Analytics:** Business users run standard SQL queries via **Amazon Athena** against the cataloged tables to generate business reports.